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Course: Advance Programming

Q1. Write a c program for three different operations as stated in question 1 to analyze the space complexity.

Ans: Given below is the ScreenShot of code and output:

```
class_assignment > C analyzeSpaceComplexity.c > linearSpace(int)
1  #include<stdio.h>
2  #include<stdlib.h>
3  void constantSpace() {
4      int a = 10, b = 20, c;
5      c = a + b;
6      printf("Constant Space Result: %d\n", c);
7  }
8  void linearSpace(int n) {
9      int *arr = (int *)malloc(n * sizeof(int));
10
11     for (int i = 0; i < n; i++) {
12         arr[i] = i;
13     }
14
15     printf("Linear Space used for %d elements\n", n);
16     free(arr);
17 }
18 void quadraticSpace(int n) {
19     int **matrix = (int **)malloc(n * sizeof(int *));
20
21     for (int i = 0; i < n; i++) {
22         matrix[i] = (int *)malloc(n * sizeof(int));
23     }
24
25     printf("Quadratic Space used for %d x %d matrix\n", n, n);
26
27     for (int i = 0; i < n; i++) {
28         free(matrix[i]);
29     }
30     free(matrix);
31 }
32
33 int main() {
34     int n;
35     printf("Enter input size: ");
36     scanf("%d", &n);
37
38     constantSpace();
39     linearSpace(n);
40     quadraticSpace(n);
41
42     return 0;
43 }
44
```

Output:

analyzeSpaceComplexity.c:36:18: error: 'n' undeclared (first use in this function)

```
scanf("%d", &n);
```

^

analyzeSpaceComplexity.c:36:18: note: each undeclared identifier is reported only once for each function it appears in

PS C:\Users\kaush\Desktop\imp\vscode\Advance programming\Assignment\class_asisgnment> cd "c:\Users\kaush\Desktop\imp\vscode\Advance programming\Assignment\class_asisgnment"

```
{ gcc analyzeSpaceComplexity.c -o analyzeSpaceComplexity } ; if ($?) { .\analyzeSpaceComplexity }
```

Enter input size: 4567

Constant Space Result: 30

Linear Space used for 4567 elements

Quadratic Space used for 4567 x 4567 matrix

PS C:\Users\kaush\Desktop\imp\vscode\Advance programming\Assignment\class_asisgnment> cd "c:\Users\kaush\Desktop\imp\vscode\Advance programming\Assignment\class_asisgnment"

```
{ gcc analyzeSpaceComplexity.c -o analyzeSpaceComplexity } ; if ($?) { .\analyzeSpaceComplexity }
```

Enter input size: 876

Constant Space Result: 30

Linear Space used for 876 elements

Quadratic Space used for 876 x 876 matrix

PS C:\Users\kaush\Desktop\imp\vscode\Advance programming\Assignment\class_asisgnment> cd "c:\Users\kaush\Desktop\imp\vscode\Advance programming\Assignment\class_asisgnment"

```
{ gcc analyzeSpaceComplexity.c -o analyzeSpaceComplexity } ; if ($?) { .\analyzeSpaceComplexity }
```

Enter input size: 345

Constant Space Result: 30

Linear Space used for 345 elements

Quadratic Space used for 345 x 345 matrix

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